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Arthur Cayley (1821–1895)

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822. On Associative Imaginaries (conclusion)

[book p. 106] In order that this may be associative, we must still have the relations

$$bg = cd,$$
$$c^2 + dg - ag - eh = 0,$$
$$d^2 + bc - ad - bh = 0,$$

which are all three of them satisfied by $g = \dfrac{cd}{b}$, $h = \dfrac{d^2 + bc - ad}{b}$, viz. we thus have the associative and commutative system

$$x^2 = ax + by,$$
$$xy = yx = cx + dy,$$
$$y^2 = \frac{cd}{b}\,x + \frac{d^2 + bc - ad}{b}\,y.$$

I did not perceive how to identify this system with any of the double algebras of B. Peirce's *Linear Associative Algebra*, see pp. 120–122 of the Reprint, *American Journal of Mathematics*, t. IV. (1881); but it has been pointed out to me by Mr C. S. Peirce that my system, in the general case $ad - bc$ not $= 0$, is expressible as a mixture of two algebras of the form (a_1^2), see p. 120 (l.c.); whereas if $ad - bc = 0$, it is reducible to the form (c_2), see p. 122 (l.c.). The object of the present Note is to exhibit in the simple case of two imaginaries the whole system of relations which must subsist between the coefficients in order that the imaginaries may be associative; that is, the system of equations which are solved implicitly by the establishment of the several multiplication tables of the memoir just referred to.

823. On the Geometrical Interpretation of Certain Formulæ in Elliptic Functions

[From the Johns Hopkins University Circulars, No. 17 (1882), p. 238.]

[book p. 107] I have given in my *Elliptic Functions* expressions for the sn^2 of $u + \frac{1}{2}K$, $u + \frac{1}{2}iK'$, $u + \frac{1}{2}K + \frac{1}{2}iK'$; but it is better to consider the dn^2, sn^2, cn^2 of these combinations respectively, and to write the formulæ thus:

$$\operatorname{dn}^2(u + \tfrac{1}{2}K) = k\,\frac{\operatorname{dn} u - (1 - k')\operatorname{sn} u \operatorname{cn} u}{\operatorname{dn} u + (1 - k')\operatorname{sn} u \operatorname{cn} u} \;=\; k\,\frac{1 - k'x - (1 - k')y}{1 - k'x + (1 - k')y},$$
$$\operatorname{sn}^2(u + \tfrac{1}{2}iK') = \frac{1}{k}\,\frac{(1 + k)\operatorname{sn} u + i\operatorname{cn} u \operatorname{dn} u}{(1 + k)\operatorname{sn} u - i\operatorname{cn} u \operatorname{dn} u} \;=\; \frac{1}{k}\,\frac{(1 + k)x + iy}{(1 + k)x - iy},$$
$$\operatorname{cn}^2\bigl(u + \tfrac{1}{2}K + \tfrac{1}{2}iK'\bigr) = -\frac{ik'}{k}\,\frac{(k - ik')\operatorname{sn} u + \operatorname{dn} u \operatorname{cn} u}{(k + ik')\operatorname{sn} u + \operatorname{dn} u \operatorname{cn} u} \;=\; -\frac{ik'}{k}\,\frac{(k - ik')x + y}{(k + ik')x + y};$$

where in the last set of values x, y are used to denote $\operatorname{sn}^2 u$ and $\operatorname{sn} u \operatorname{cn} u \operatorname{dn} u$ respectively; and the formulæ are thus brought into connexion with the cubic curve $y^2 = x(1 - x)(1 - k^2x)$. The curve has an inflexion at infinity on the line $x = 0$; and the three tangents from the inflexion are $x = 0$, $x = 1$, $x = \dfrac{1}{k^2}$, touching the curve at the points x, $y = (0, 0)$, $(1, 0)$, $\left(\dfrac{1}{k^2}, 0\right)$ respectively; hence these points are sextactic points.

We may from any one of them, for instance the point $(0, 0)$, draw four tangents to the curve, $(1 + k)x + iy = 0$, $(1 + k)x - iy = 0$; $(1 - k)x + iy = 0$, $(1 - k)x - iy = 0$; where the first and second of these lines form a pair, and the third and fourth of them form a pair, viz. the two tangents of a pair touch in points such that the line joining them passes through the point of inflexion: in particular, for the first-mentioned pair, the equation of the line joining the points of contact is $\frac{1}{1+k}x = 0$. The linear functions belonging to a pair of tangents are precisely those which present themselves in the formulæ; thus if $2X = (1+k)x + iy$, $2Y = (1+k)x - iy$, the second of the three formulæ is $\mathrm{sn}^2(u + \frac{1}{2}iK') = \frac{1}{k}\frac{X}{Y}$; and the other two formulæ correspond in like manner to pairs of

[book p. 108] tangents from the sextactic points $\left(\frac{1}{k^2}, 0\right)$ and $(1, 0)$ respectively. The formulæ are connected with the fundamental equations expressing the functions sn, cn, dn as quotients of theta functions.

824. Note on the Formulæ of Trigonometry

[From the Johns Hopkins University Circulars, No. 17 (1882), p. 241.]

[book p. 108] The equations $a = c\cos B + b\cos C$, $b = a\cos C + c\cos A$, $c = b\cos A + a\cos B$, which connect together the sides a, b, c and the angles A, B, C of a plane triangle, may be presented in an algebraical rational form, by introducing in place of the angles A, B, C the functions $\cos A + i\sin A$, $\cos B + i\sin B$, $\cos C + i\sin C$, viz. calling these x, y, z respectively, or, what is the same thing, writing

$$2\cos A = \frac{x}{w} + \frac{w}{x},\quad 2\cos B = \frac{y}{w} + \frac{w}{y},\quad 2\cos C = \frac{z}{w} + \frac{w}{z},$$

then the foregoing equations may be written

$$\begin{aligned}
&(-\,2yzw\,,\ y(z^2+w^2)\,,\ z(y^2+w^2)\,|\,a\,,\,b\,,\,c) = 0,\\
&(z(x^2+w^2)\,,\ -\,2zxw\,,\ x(z^2+w^2)\,|\,a\,,\,b\,,\,c) = 0,\\
&(y(x^2+w^2)\,,\ x(y^2+w^2)\,,\ -\,2xyw\,|\,a\,,\,b\,,\,c) = 0;
\end{aligned}$$

that is, as a system of bipartite equations linear in (a, b, c) and cubic in (x, y, z, w) respectively.

Similarly in Spherical Trigonometry, writing as above for the angles, and for the sides writing in like manner $2\cos a = \frac{\alpha}{\delta} + \frac{\delta}{\alpha}$, $2\cos b = \frac{\beta}{\delta} + \frac{\delta}{\beta}$, $2\cos c = \frac{\gamma}{\delta} + \frac{\delta}{\gamma}$, we have a system of bipartite equations separately homogeneous in regard to (x, y, z, w) and $(\alpha, \beta, \gamma, \delta)$ respectively.

825. A Memoir on the Abelian and Theta Functions

[Chapters I to III, American Journal of Mathematics, t. V. (1882), pp. 137–179; Chapters IV to VII, ib., t. VII. (1885), pp. 101–167.]

[book p. 109] The present memoir is based upon Clebsch and Gordan's *Theorie der Abel'schen Functionen*, Leipzig, 1866 (here cited as C. and G.); the employment of differential rather than of integral equations is a novelty; but the chief addition to the theory consists in the determination which I have made for the cubic curve, and also (but not as yet in a perfect form) for the quartic curve, of the differential expression $d\Pi^{1\alpha}_{1\beta}$ (or as I write it $d\Pi_{12}$) in the integral of the third kind $\int_a^b d\Pi^{1\alpha}_{1\beta}$ in the final normal form (*endliche Normalform*) for which we have (p. 117) $d\Pi^{1\alpha}_{a\beta} = \int_a^b d\Pi^{a\beta}_{1\beta}$, the limits and parametric points interchangeable. The want

of this determination presented itself to me as a lacuna in the theory during the course of lectures on the subject which I had the pleasure of giving at the Johns Hopkins University, Baltimore, U.S.A., in the months January to June, 1882, and I succeeded in effecting it for the cubic curve; but it was not until shortly after my return to England that I was able partially to effect the like determination in the far more difficult case of the quartic curve. The memoir contains, with additional developments, a reproduction of the course of lectures just referred to. I have endeavoured to simplify as much as possible the notations and demonstrations of Clebsch and Gordan's admirable treatise; to bring some of the geometrical results into greater prominence; and to illustrate the theory by examples in regard to the cubic, the nodal quartic, and the general quartic curves respectively. The various chapters are: I, Abel's Theorem; II, Proof of Abel's Theorem; III, The Major Function; IV, The Major Function (continued); V, Miscellaneous Investigations; VI, The Nodal Quartic; VII, The Functions T, U, V, Θ. The paragraphs of the whole memoir will be numbered continuously.

Chapter I. Abel's Theorem.

[book p. 110] The Differential Pure and Affected Theorems. Art. Nos. 1 to 5.

1. We have a fixed curve and a variable curve, and the differential pure theorem consists in a set of linear relations between the displacements of the intersections of the two curves; in the affected theorem, a linear function of the displacements is equated to another differential expression. I state the two theorems, giving afterwards the necessary explanations.

The pure theorem is

$$\Sigma\,(x,\,y,\,z)^{n-3}\,d\omega = 0.$$

The affected theorem is

$$\Sigma\,(x,\,y,\,z)^{n-3}\,\frac{\delta\phi}{J(\phi,\,f)}\,d\omega = -\,\frac{\delta\phi_1}{\phi_1}\cdot\frac{\delta\phi_2}{\phi_2}.$$

2. We have a fixed curve $f = 0$, or say the curve f, or simply the fixed curve, of the order n, with δ dps, and therefore of the deficiency $\frac{1}{2}(n-1)(n-2) - \delta, = p$. The expression "the dps" means always the δ dps of f.

And we have a variable curve $\phi = 0$, or say the curve ϕ, or simply the variable curve, of the order m, passing through the dps and besides meeting the fixed curve in $mn - 2\delta$ variable points.

Moreover, $d\omega$ is the displacement of the current point 0, coordinates $(x,\,y,\,z)$, on the fixed curve, viz. the equation $f = 0$ gives

$$\frac{df}{dx}\,dx + \frac{df}{dy}\,dy + \frac{df}{dz}\,dz = 0,$$

and we thence have

$$\frac{df}{dx} : \frac{df}{dy} : \frac{df}{dz} \;=\; y\,dz - z\,dy \,:\, z\,dx - x\,dz \,:\, x\,dy - y\,dx,$$

so that we have three equal values each of which is put $= d\omega$, viz. we write

$$\frac{y\,dz - z\,dy}{\dfrac{df}{dx}} = \frac{z\,dx - x\,dz}{\dfrac{df}{dy}} = \frac{x\,dy - y\,dx}{\dfrac{df}{dz}} = d\omega,$$

and the ω thus defined is the displacement.

Note. For comparison with C. and G., we observe that in the equation, p. 47, $C = \frac{p\,dq - q\,dp}{f'_r}$ (p, q, r belong to the upper limit and corresponds to my ϕ; the equation gives therefore $C = -\dfrac{\delta\phi}{J}\,d\omega$, agreeing with the formula in the text.
[book p. 111] $(x, y, z)^{n-3} = 0$ is the minor curve, viz. the general curve of the order $n - 3$, which passes through the dps*, and the function $(x, y, z)^{n-3}$ is the minor function.

1. and **2.** fixed points at 0, of the parametric points, coordinates (x_1, y_1, z_1) and (x_2, y_2, z_2) respectively; and 012 denotes the determinant

$$012 = \begin{vmatrix} x & y & z \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix},$$

so that $012 = 0$ is the equation of the line joining the points 1 and 2: this line meets the fixed curve in $n - 2$ other points, called the residues of 1, 2.

$(x, y, z)^{n-3} = 0$ is the major curve passed the points 1 and 2; viz. this is the general curve of the order $n - 3$, passing through the dps and also through the residues of 1, 2.

But further, the function $(x, y, z)^{n-3}$ is the proper major function; viz. the implicit factor of the function is as determined that, taking $0 = 1$, the current point at 1, that is, writing (x, y, z) for (x_1, y_1, z_1), the function $(x, y, z)^{n-3}$ reduces itself to the polar function $\big(x_2\frac{d}{dx} + y_2\frac{d}{dy} + z_2\frac{d}{dz}\big)\phi_1$, afterwards written $n.\,1^{n-2}\,2$; this implies that taking $0 = 2$, the current point at 2, the function reduces itself to the polar function $n.\,12^{n-2}$.

2. ϕ is what ϕ becomes at writing therein (x_1, y_1, z_1) for (x, y, z); and similarly ϕ_2 is what ϕ becomes on writing therein (x_2, y_2, z_2) for (x, y, z).

δ denotes differentiation in regard only to the coefficients of ϕ, viz. writing $\phi = (a, b, c, \ldots)(x, y, z)^m$ have $\delta\phi = (\delta a, \delta b, \delta c, \ldots)(x, y, z)^m$, and similarly $\delta\phi_1 = (\delta a, \delta b, \delta c, \ldots)(x_1, y_1, z_1)^m$ and $\delta\phi_2 = (\delta a, \delta b, \delta c, \ldots)(x_2, y_2, z_2)^m$.

The sum Σ extends to all the variable intersections of the two curves.

3. As to the meaning of the theorem, consider first the pure theorem. The variable intersections are not all of them arbitrary points on the fixed curve; a certain number of them taken at pleasure on the fixed curve will determine the remaining variable intersections; and there are thus in fact, between the displacements of all of them, 2δ equations of integral relation; in such a way that there only remains a number of integral displacement-relations equal to p, the deficiency of the curve.

But from the values of L_1, L_2 we have $x_3 = \frac{1}{2}L_2 + L_1$, $x_1 + x_2 = \frac{1}{2}L_2 + L_1$; and the coefficient of L_1L_2 in $L_2L_3 + L_2L_1 + L_1L_2$ is the equation is thus verified.

The example would perhaps have been more instructive if the working out would have been more difficult.

* This theory was demonstrated in my Section A of the British Association at the York meeting, Sept. 3, 1881, Report (London), 1881, [712], 'On the simplification of the equation given therefor in Cayley's Mem. "A Partial Differential Equation associated with the simplest case of Abel's Theorem." Vol. XII.
[book p. 112] **4.** It is of course important to show, and it will be shown, that the number of independent displacement-relations given by the theorem is equal to the number of independent integral relations between the variable intersections.

5. Observe that the pure theorem gives all the displacement-relations between the variable intersections; we are hereby led to see the nature of the affected theorem, viz. in this same case where the variable curve ϕ meets the fixed curve in such a way as to produce variations $\delta\phi_1$, $\delta\phi_2$ in ϕ_1, ϕ_2, the displacement-relations $\Sigma\,(x, y, z)^{n-3}\,d\omega = 0$ given by the pure theorem.

Examples of the Pure Theorem—The Fixed Curve a Circle. **Art. Nos. 6 to 13.**

6. The pure theorem is not applicable to the case $n = 2$, the fixed curve a conic: it is in fact gives no displacement-relations, and this is as it should be, for the variable intersections are all of them arbitrary.

The next case is $n = 3$, $\delta = 0$, the fixed curve a cubic. For greater simplicity the equation is taken in Cartesian coordinates. In general for such an equation, writing in the homogeneous formula $z = 1$, we have

$$d\omega = \frac{dy}{\frac{df}{dx}} = -\frac{dx}{\frac{df}{dy}};$$

the two values being of course equal in virtue of $\frac{df}{dx}\,dx + \frac{df}{dy}\,dy = 0$; taking the former value and considering $\frac{df}{dy}$ as expressed in terms of x, let this be called P (of course, P is an irrational function of x): then we have $d\omega = \frac{dx}{P}$, and similarly $d\omega_1 = \frac{dx_1}{P_1}$, &c.

The fixed curve being then a cubic, let the variable curve be a line; this meets the cubic in three points, say 1, 2, 3; and any two of these determine the third, the relation being $x_1+x_2+x_3 =$ (a constant). In Cartesian coordinates, if the equation of the line be $\alpha x + \beta y + \gamma = 0$, then writing for y its value in terms of x, the equation of the cubic becomes

$$f = -\frac{1}{\beta^3}\,(\alpha x + \gamma)^3 + a(\alpha x + \gamma)^2 + b(\alpha x + \gamma) + c.$$

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* The coefficients a and b are determined by means of the points 3 and 4, that is, they are functions of x_1, x_2, considering them as those expressed, thus determined.

[book p. 113] The corresponding integral equation is the equation which expresses that the points 1, 2, 3 are in a line, viz. considering y_1, y_2, y_3 as given functions of x_1, x_2, x_3 respectively; this is

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0,$$

or, in the notation already made use of for such a determinant, $123 = 0$.

7. The equation $\Sigma\, d\omega = 0$, where the direction $\frac{dx}{P}$ has a peculiar interpretation when we consider the coefficients of the cubic as arbitrary constants, and therefore the cubic as a moving curve in a moving changing also arbitrary changing; but the result then can be obtained by elimination from $\Sigma\,\frac{dx}{P} = 0$ and the equation $123 = 0$. There is in fact a theorem of intuitive interpretation, $dx + dx_3 + dx_3 = 0$, we have thus

$$\frac{dx_1}{P_1} + \frac{dx_2}{P_2} + \frac{dx_3}{P_3} = 0,$$

and we hence see that, for $\tau = 0$,

$$\frac{dx_1}{P_1} = \frac{dx_2}{P_2};$$

an equation which will be useful.

The Preparation for the Affected Theorem resumed. **Art. No. 27.**

27. In the affected theorem instead of $(x,\ y,\ z)$ we introduce the new coordinates $(\rho,\ \sigma,\ \tau)$. We have

$$J(\phi,\ f,\ \tau)=\frac{d(\phi,\ f,\ \tau)}{d(\rho,\ \sigma,\ \tau)}\cdot\frac{d(\rho,\ \sigma,\ \tau)}{d(x,\ y,\ z)},$$

where the first factor is $=\dfrac{d(\phi,\ f)}{d(\rho,\ \sigma)}$, say that this is $J(\phi,\ f)$, the Jacobian in regard to ρ, σ: and the second factor is at once found to be $=\Delta!$. We have consequently

$$\frac{1}{J(\phi,\ f,\ \tau)}=\frac{1}{\Delta!\,J(\phi,\ f)},$$

and the equation for the affected theorem becomes

$$\Sigma\,(x,\ y,\ z)^{n-3}\,\frac{\delta\phi}{J(\phi,\ f)}=-\,\Delta!\,\left(-\,\frac{\delta\phi_1}{\phi_1}\cdot\frac{\delta\phi_2}{\phi_2}\right),$$

where $(x,\ y,\ z)^{n-3}$ is to be regarded as standing for its value in terms of $(\rho,\ \sigma,\ \tau)$.

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* This theorem was communicated to me in Section A of the British Association at the York meeting, Sept. 3, 1881, [712]. 'On a Partial Differential Equation associated with the simplest case of Abel's Theorem.' Vol. XII.

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[book p. 114] **8.** As a particular example, let the fixed curve be the cubic $y^2=x(1-x)(1-k^2x)$, and the line $x=b$; the integral relation $123=0$. I give a direct verification. To find x_3, the coefficient of the cubic being on the part of the relations of the cubic, 1, 2, 3, we have, by definition, $x_1\,x_2\,x_3$ as the roots of equation $0=\dfrac{1}{k^2}\,(c+by)^2-x(1-x)(1-k^2x)$.

The fact, x_3 depends in a like manner on x_1, x_2, and the integral relation $123=0$. I give a direct verification. To find x_3, the cubic being substituted for each of the cubic terms in [] are equal, that is, we ought to have

$$y_3(x_1-x_2)=y_1(x_3-x_2)-y_2(x_3-x_1),\quad\text{(say A.)}.$$

Comparing for instance the first and second terms, the equation is

$$-\,y_3(x_3-x_1)+y_1(x_2-x_3)+y_2(x_3-x_1)+y_3(x_1-x_2)=0;$$

or, as this may be written,

$$-\,(x_1+x_3)y_3+x_3y_1+(x_1+x_2)y_2+x_2(y_3-y_1)=0;$$

where the whole condition for the line $x=b$ being substituted, the coefficients K may be determined.

[book p. 115] $d\omega_3+d\omega_3+d\omega_3=0$, and the integral relation $123=0$. This last equation is an integral of the differential equation $d\omega_1+d\omega_2+d\omega_3=0$, not containing any arbitrary constant, it is a particular integral.

But regard one of the three points, say 3, as a fixed point, that is, let the line pass through the fixed point 3 of the cubic, and besides meet it in the variable points 1 and 2. We write $d\omega_3=0$, and the differential equation thus is $d\omega_1+d\omega_2=0$, while the integral equation is now $123=0$, the general integral of $d\omega_1+d\omega_2=0$.

Let the variable curve be a conic; say the intersections with the cubic are 1, 2, 3, 4, 5, 6. Any three of these points determine the conic, and therefore the two remaining intersections; that is, the points 4, 5, 6 are integral relations, and consequently between the differential equations apparently $d\omega_1+d\omega_2+\ldots+d\omega_6=0$.

We have therefore the cubic 123456 = 0, say a particular integral of $d\omega_1+d\omega_2+\ldots+d\omega_6=0$.

If, however, we take 3 a fixed point on the cubic, we then have the same equation as the general integral of $d\omega_1+d\omega_2+\ldots+d\omega_5=0$, which expresses the determination, with the same line on the conic.

But taking also 3 a fixed point of the cubic, we have $d\omega_3=0$; and there should therefore be one integral relation, the equation 123456 = 0, which expresses the determination on the conic.

We have thus 123456 = 0 as a particular integral of

$$d\omega_1+d\omega_2+d\omega_3+d\omega_4+d\omega_5+d\omega_6=0.$$

If, however, we take 3 a fixed point on the cubic, we then have the same equation as the general integral of $d\omega_1+d\omega_2+d\omega_3+d\omega_4+d\omega_5=0$, which expresses the determination, with the same line on the conic.

But taking also 3 a fixed point of the cubic, we have $d\omega_3=0$; and there should therefore be one integral relation between $d\omega_1$, $d\omega_2$, $d\omega_4$, $d\omega_5$, $d\omega_6$, that is, the same equation as the integral of $d\omega_1+d\omega_2+d\omega_4+d\omega_5+d\omega_6=0$.

* The symbol explains, I think, itself; the point may be any point on the plane (so it is some constant which remains arbitrary but which disappears from the difference $-\dfrac{x_1\,x_2}{P}$; this = 0 is explained further on in the text.

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[book p. 116] **9.** determinants point, independent of the particular conic through 4, 5 and 6 used for the construction. Thus 4 and 5 being any two chosen points on the cubic, the residual point 6 is, in fact, the intersection with the cubic of the line through 4 and 5.

12. We might go on to the case where the variable curve is a cubic: this meets the fixed curve in 9 points, but they do not determine the variable cubic; for, the cubic being assumed determined by 9 free constants, and corresponding to it the equation gives therefore $\dfrac{d\omega_1}{P_1}+\ldots+\dfrac{d\omega_9}{P_9}=0$.

Examples of the Pure Theorem—Fixed Curve a Circle. *Art. Nos. 13 to 14.*

13. The fixed curve is taken to be the circle $x^2+y^2=1$, and the parametric points 1 and 2 to be the points (1, 0) and (0, 1) on this circle. The variable curve is a conic

$$\Sigma\begin{vmatrix} a, & y, & 1\\ 1, & 0, & 1\\ 0, & 1, & 1\end{vmatrix}=0,$$

have 012 denoting the determinant.

Starting from the formula

$$\Sigma\,\frac{(x,\,y)\,(x,\,y_1,\,1)\,d\omega}{012}=-\,\frac{\delta\phi_1}{\phi_1}\cdot\frac{\delta\phi_2}{\phi_2},$$

where the summation extends to the points 3 and 4, viz. if P_1 be a constant, = 2, (if, that is, $\sin u=a$, $2(\cos u+a)=b$ in the place of u, the points 3 and 4 are at the distances $a+b$ and $a-b$, and therefore equal); say $\theta=u+\xi$, $\theta=u-\xi$; we have $\sin u=a$, and so

$$\frac{x_3+x_4}{2\sin u}=\frac{a+b}{2\sin u},\quad \frac{x_3-x_4}{2\sin u}=\frac{a-b}{2\sin u};$$

and similarly

$$\frac{y_3}{y_4}=\frac{dx_3}{dx_4},\quad \frac{dx_3}{dy_4}=\frac{dx_4}{dy_3}.$$

The formula

$$\Sigma \frac{1}{y(x+y-1)} = -\frac{\delta\phi_1}{\phi_1} \cdot \frac{\delta\phi_2}{\phi_2}$$

becomes

$$2 \frac{1}{y(x+y-1)} = -\frac{\delta\phi_1}{\phi_1} \cdot \frac{\delta\phi_2}{\phi_2}.$$

The coefficients a and b are determined by means of the points 3 and 4, that is, they are functions of x_1, x_2, considering them as those expressed, thus considered.

[book p. 117] **14.** Writing P, Q, $R = y-p$, x_1+x-q, $-x-y+r$, also x_3, y_3, the displacements $\dfrac{dx_3}{p_3}$ and $\dfrac{dx_4}{p_4}$ are the two arbitrary points 3 and 4 on the conic which equation must become an identity in regard to the terms in dx_3, dx_4 respectively. It only remains to verify that this is so.

15. Writing P, Q, $R = y-p$, x_1+x-q, $-x-y+r$, also x_3, y_3, $-x_3-y_3+r$ respectively, $-x_4-y_4+r$ for the values of P, Q, R at the points 1, 2, 3, 4, we have

$$dP \cdot dQ \cdot dR \cdot \frac{1}{e(x-p)(x_3-p)(x_4-p)} \big[(Q'-R)\,dP + (R'-P)\,dQ + (P'-Q)\,dR\big] = 0,$$

and with these values, and by aid of the relations

$$Q-R,\ R-P,\ P-Q,\ P'-R',\ R'-P',\ P'-Q' = -P+R-L,$$

the equation is found to be

$$\frac{dx_3}{p_3} + \frac{dx_4}{p_4} = \frac{(L_3(x_4-p)+L_4(x_3-p))}{(x_4-p)(p_3-p_4)\dfrac{e}{(x_3-p)}} \left(\frac{dq_1}{q_1} - \frac{dq_2}{q_2}\right);$$

viz. this will be true if only

$$L_3(x_4-p) + L_4(x_3-p) = 0\,\frac{e}{(x_3-p)\,(x_4-p)};$$

that is,

$$-x_3\,P_3' - x_4\,P_4' + L_3(x_4-p) + L_4(x_3-p) = 0.$$

But from the values of L_1, L_2, we have $-x_3 = \frac{1}{2}L_3 + L_4$, $x_3+x_4 = \frac{1}{2}L_4 + L_3$; and the coefficient of L_4 in $L_3L_4 + L_4L_3 + L_3L_4$ is the equation is thus verified.

The example would perhaps have been more instructive if the working out would have been more difficult.

The Variable Intersections of the Two Curves—Number of Independent Integral Relations. **Art. Nos. 15 to 19.**

15. Suppose $n=3$, $\delta=0$ $(p=1)$; the fixed curve is a cubic; and suppose successively $m=1$, 2, 3, ..., the variable curve a line, conic, cubic, &c.

If $m=1$, then two points on the cubic determine the line, and consequently the remaining intersection with the cubic; hence there are two integral relations.

If $m=2$, then five points on the cubic determine the conic, and therefore the remaining intersection; hence there is one integral relation.

[book p. 118] If $m=3$, then nine points on the cubic determine the cubic; hence there are no integral relations. The result is consistent with the variable curve having $m^2 = m^2$ disposable constants, $\frac{1}{2}m(m+3)$, and the variable curve passing through $mn = 3m$ points on the fixed curve, we have $\frac{1}{2}m(m+3) - 3m = \frac{1}{2}(m-1)(m-2)$, $= p$, that is the number of integral relations is the deficiency of the cubic.

16. Suppose next $n = 4$, $\delta = 0$ ($p = 3$); the fixed curve a general quartic; and so before suppose successively $m = 1, 2, 3, \ldots$, the variable curve a line, conic, cubic, &c.

If $m = 1$, then three points on the quartic determine the line, and therefore the remaining two intersections; the number of integral relations is $= 3$.

If $m = 2$, then five points on the quartic determine the conic, and therefore the remaining three intersections; the number of integral relations is $= 3$.

If $m = 3$, then nine points on the quartic determine the cubic, and therefore the remaining three intersections; the number of integral relations is $= 3$.

If $m = 4$, then fourteen points on the quartic determine the quartic, and therefore the remaining two intersections; the number of integral relations is $= 2$.

If $m = 5$, then twenty points on the quartic determine the quartic, and therefore there is one integral relation; the number of integral relations is $= 1$.

The number of intersections is $4 = mn, = 4m$; the variable curve has $\frac{1}{2}m(m+3)$ disposable constants; hence the number of integral relations is in general the larger of the two numbers

$$m^2 - 4m + 1 \left(= \tfrac{1}{2}(m-1)(m-2) + (m-2)(m-3)\right),$$

$= \frac{1}{2}(m-1)(m-2) + 2(m-3)$, that is, the number of integral relations is, as we presently see, $= p$.

The integral equations consist of throughout out of course independent relations.

The Variable Intersections of the Two Curves. Number of Independent Displacement Relations given by the Pure Theorem. *Art. No. 20.*

20. The number of displacement-relations given by the pure theorem is a number of constants in minor function $(x, y, z)^{n-3}$, which equated to zero represents a curve passing through the dps, this is always $= p$.

[book p. 119] **18.** The reasoning is utterly altered in the case where the fixed curve has dps; then considering the general case of a fixed curve n, δ, p,—

If $m = 1$, then $2 - \delta$ points on the fixed curve (this implies $\delta \not> 2$) determine the line, and therefore the remaining $n - 2\delta - 2, = n - 2 - 2$ intersections; the number of integral relations is $= n - 2$.

If $m = 2$, then $5 - 2\delta$ points on the conic determine the conic, and therefore the remaining $2n - 2\delta - 5, = 2n - 5$ intersections; the number of integral relations is $= 2n - 5$.

If $m = 3$, then $9 - 3\delta$ points on the cubic determine the cubic, and therefore the remaining $3n - 2\delta - 9, = 3n - 9$ intersections; the number of integral relations is $= 3n - 9$.

If $m = n-3$, then $\frac{1}{2}(n-3)n - \delta(n-3)$ points on the curve determine the curve, and therefore the remaining $(n-3)n - 2\delta - \frac{1}{2}(n-3)n, = \frac{1}{2}(n-3)n - 2\delta$, intersections; the number of integral relations is $= \frac{1}{2}(n-3)n - 2\delta$. Hence the number of integral relations in cases $m = n-3, = p$.

If we have, then $\frac{1}{2}n - 1)(n+4) - \delta$ points on the curve determine the curve, and therefore $1 - \delta$ intersections (it should be $= p$).

The integral equations consist of throughout out of course independent relations.

The Variable Intersections of the Two Curves—Number of Independent Displacement Relations given by the Pure Theorem. *Art. No. 20.*

20. The number of displacement-relations given by the pure theorem is a number of constants in minor function $\phi(x, y, z)^{n-3}$, which equated to zero represents a curve passing through the dps, this is always $= p$. Consider three consecutive positions of the line, meeting the cubic in the points 1, 2, 3; $1'$, $2'$, $3'$; and $1''$, $2''$, $3''$ respectively. By $x = 0$ is satisfied an identity. Hence the number of integral displacement-relations is $= 1, = p$.

But it remains to verify, viz. if U be any rational and integral function $(x, y, z)^{n-3}$, then we have

$$\Sigma\,(x,\, y,\, z)^{n-3}\, d\omega = 0, \quad p,\ 0,\ \sigma,\ \tau;\ \eta\,(2x,\, 1)\,\phi\, 0,$$

or we might also use the symbolic form

$$\Sigma\,(\rho 1+\sigma 2+\tau 3)^n=0.$$

So for $n=4$, $\delta=0$, $m=1$, the displacement-relations are

$$\Sigma\,(x,\,y,\,z)\,d\omega=0,\ \text{that is,}\ \Sigma\,x\,d\omega=0,\ \Sigma\,y\,d\omega=0,\ \Sigma\,z\,d\omega=0;$$

which are 3 in number, $=p$.
[book p. 120] and then six equations give conversely $\Sigma\,(x,\,y,\,z)\,d\omega=0$, and in particular they give $\Sigma\,(x,\,y,\,z)\,d\omega=0$. Thus, if at pleasure p, q, r at most $=n-3$, the variable curve has $\frac{1}{2}n(n+3)-2\delta$ disposable constants; hence the number of integral displacement-relations is in general

$$m_p-p\,(m+1)\,;\quad \text{and the other two formulas correspond in like manner.}$$

since in this case the relations given by the theorem are all $=0$, but each contained p relations independent. It is easily seen that the number is in general

$$m_p=\tfrac{1}{2}n(n+3)-2\delta-(m-1)(2m-1),\quad \text{the number of independent displacement relations}=3.$$

The reasoning is utterly altered in the case where the fixed curve has dps; we have generally

$$m\,p=p,\ \text{or as in case}\ n=4,\ \delta=0;$$

while the number of integral relations $=p$.

Hence the difference $\frac{1}{2}n(n+3)-2\delta-(m-1)(2m-1)$, which is equal to $\frac{1}{2}(m-1)(m-2)+\frac{1}{2}m(m-3)$, $=p$, $=p$, the number of independent displacement-relations $=3$.

So for $n=4$, $\delta=0$, $m=1$, the displacement-relations are $\Sigma\,(x,\,y,\,z)\,d\omega=0$, that is, $\Sigma\,x\,d\omega=0$, $\Sigma\,y\,d\omega=0$, $\Sigma\,z\,d\omega=0$, in all three, $=p$.

As to the dps of the Fixed Curve. *Art. No. 21.*

21. I conclude with a general remark applicable to the whole of the three chapters. There is no necessity to attend to all or indeed to any of the dps of the fixed curve. For although we may pass through all of them altogether (as e.g. if they are nodes by taking the variable curve a multiple-curve passing through them all and treating each dp as a multiple point), the number which presents itself is not the number of dps of the fixed curve, but the difference between the dps of the fixed curve and the number which it is necessary to attend to in any of the simplifications introduced therein by taking advantage thereof.

Chapter II. Proof of Abel's Theorem.

Preparation. Art. Nos. 22 and 23.

22. Starting from the equation $\phi=(a,\,b,\,c,\ldots)(x,\,y,\,z)^m=0$ of the variable curve, we have

$$\frac{d\phi}{dx}\,dx+\frac{d\phi}{dy}\,dy+\frac{d\phi}{dz}\,dz=0,$$

$$\frac{d\phi}{dx}=\frac{d\phi}{dx_1}\,\frac{dx_1}{dx}+\frac{d\phi}{dz_1}\,\frac{dz_1}{dx},\quad \&c.;$$

[book p. 121] where $\delta\phi=(\delta a,\,\delta b,\,\delta c,\ldots)(x,\,y,\,z)^m$. Let τ denote an arbitrary linear function, $\omega=xdy+\ldots$; multiply the two equations by $dy+\omega$ respectively, and add: the coefficient of δa

is $-(xdy - ydx) + \omega(xdz - zdx) + \tau(xdy - ydx), = -d\tau$, say; and so on; we thus arrive at the equation

$$(ydx - zdy)\frac{d\phi}{dx} + (zdx - xdz)\frac{d\phi}{dy} + (xdy - ydx)\frac{d\phi}{dz} + \delta\phi = 0,$$

introducing $d\omega$, this becomes

$$d\omega\left[\left(\frac{df}{dy}\frac{d\phi}{dz} - \frac{df}{dz}\frac{d\phi}{dy}\right) + \omega\left(\frac{df}{dx}\frac{d\phi}{dz} - \frac{df}{dz}\frac{d\phi}{dx}\right) + \tau\left(\frac{df}{dx}\frac{d\phi}{dy} - \frac{df}{dy}\frac{d\phi}{dx}\right)\right] + \delta\phi = 0;$$

or observing that ω, τ, σ are the differential coefficients $\frac{d\sigma}{dx}$, $\frac{d\sigma}{dy}$, $\frac{d\sigma}{dz}$, the term in [] is $J(\tau,\ f,\ \phi)$, or say $J(\phi,\ f,\ \tau)$, where

$$J(\phi,\ f,\ \tau) = \frac{d(\phi,\ f,\ \tau)}{d(x,\ y,\ z)},$$

and we have hence

$$d\omega = \frac{-\delta\phi}{J(\phi,\ f,\ \tau)}.$$

23. The two theorems thus become

$$\Sigma\,(x,\ y,\ z)^{n-3}\,\frac{-\delta\phi}{J(\phi,\ f,\ \tau)} = 0;$$

$$\Sigma\,(x,\ y,\ z)^{n-3}\,\frac{-\delta\phi}{J(\phi,\ f,\ \tau)} = -\frac{\delta\phi_1}{\phi_1}\cdot\frac{\delta\phi_2}{\phi_2}.$$

But further, if in the first equation we write $\tau = z$, and in the second equation we retain τ, using it then to denote the linear function 012, we have the equations

$$\Sigma\,(x,\ y,\ z)^{n-3}\,\frac{\delta\phi}{J(\phi,\ f)} = 0,$$

$$\Sigma\,(x,\ y,\ z)^{n-3}\,\frac{\delta\phi}{J(\phi,\ f,\ 012)} = -\frac{\delta\phi_1}{\phi_1}\cdot\frac{\delta\phi_2}{\phi_2}.$$

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[book p. 122] where, in the first equation, $J(\phi,\ f)$ denotes the Jacobian

$$J(\phi,\ f) = \begin{vmatrix} \frac{d\phi}{dx} & \frac{d\phi}{dy} \\ \frac{df}{dx} & \frac{df}{dy} \end{vmatrix},\quad = \frac{d(\phi,\ f)}{d(x,\ y)}.$$

In these equations the only differential symbol is the δ affecting the coefficients a, b,... of ϕ, viz. the equations are, in regard to such coefficients, algebraical equations, which are in fact given by Jacobi's Fraction-theorem, as I shall explain. But for the further reduction of the affected theorem I interpose the next chapter.

The Coordinates **$(\rho,\ \sigma,\ \tau)$.** *Art. Nos. 24 to 26.*

24. The letter τ has just been used to denote the determinant 012: there is often occasion to consider three points 1, 2, 3 coordinates $(x_1,\ y_1,\ z_1)$, $(x_2,\ y_2,\ z_2)$, $(x_3,\ y_3,\ z_3)$ respectively; and then writing ρ, σ, τ as the determinants 023, 031, 012 respectively, and Δ as the determinant 123, we have identically

$$\Delta x = x_1\rho + x_2\sigma + x_3\tau,$$

$$\Delta y = y_1\rho + y_2\sigma + y_3\tau,$$

$$\Delta z = z_1\rho + z_2\sigma + z_3\tau,$$

which equations, regarding therein the point 0, coordinates $(x,\, y,\, z)$, as a current point, are in fact equations for transformation from the coordinates $(x,\, y,\, z)$ to the coordinates $\rho,\, \sigma,\, \tau$, belonging to the triangle of reference 123. The points 1 and 2 have been already taken to be on the fixed curve, and it will be assumed that 3 is also a point on this curve.

25. If the function f, which equated to zero gives the equation of the fixed curve, be transformed to the new coordinates $(\rho,\, \sigma,\, \tau)$, the coefficients of the transformed function are polar functions, each divided by ∇^n, viz. the coefficient of ρ^n is $\dfrac{1}{\nabla^n}\, 1^n$, which by reason that 1 is a point on the curve is $= 0$ (and similarly the coefficients of σ^n and of τ^n are each $= 0$); the coefficient of $\rho^{n-1}\sigma$ is $\dfrac{1}{\nabla^n}\, n.\, 1^{n-1}\, 2$; that of $\rho^{n-1}\tau$ is $\dfrac{1}{\nabla^n}\, n.\, 1^{n-1}\, 3$; that of $\rho^{n-2}\sigma^2$ is $\dfrac{1}{\nabla^n}\, \tfrac{1}{2}n(n-1).\, 1^{n-2}\, 2^2$; and so in other cases. I write this in the form

$$f = \frac{1}{\nabla^n}\,(1^n,\, \dots\ )(\rho,\, \sigma,\, \tau)^n;$$

or we might also use the symbolic form

$$f = \frac{1}{\nabla^n}\,(\rho 1 + \sigma 2 + \tau 3)^n.$$

[book p. 123] The terms independent of τ contain, it is clear, the factor $\rho\sigma$, and separating these terms from the others, the equation may be written

$$f = \frac{1}{\nabla^n}\,\rho\sigma\,(n.\, 1^{n-1}\, 2,\, \dots\ )(\rho,\, \sigma)^{n-2} + \text{terms involving } \tau.$$

26. The equations $\tau = 0$, $(\dots\ )(\rho,\, \sigma)^{n-2} = 0$, determine the residues of the points 1, 2, and hence the major function $(x,\, y,\, z)^{n-3}$, expressed in terms of $\rho,\, \sigma,\, \tau$, and writing therein $\tau = 0$, must reduce itself to $(\dots\ )(\rho,\, \sigma)^{n-2}$ multiplied by a constant factor which can be found to be $= \dfrac{1}{\nabla^{n-1}}$: for taking the current point at 1 we have $(\rho,\, \sigma,\, \tau) = (\Delta,\, 0,\, 0)$, and the corresponding value of the major function is thus equal, and in this case $\dfrac{1}{\nabla^{n-1}}\, n.\, 1^{n-1}\, 2$, $= n.\, 1^{n-1}\, 2$, as it ought to be. We have thus

$$(x,\, y,\, z)^{n-3} = \frac{1}{\nabla^{n-1}}\,(n.\, 1^{n-1}\, 2,\, \dots\ )(\rho,\, \sigma)^{n-2} + \text{terms involving } \tau;$$

and we hence see that, for $\tau = 0$,

$$(x,\, y,\, z)^{n-3} = \frac{\Delta f}{\rho\sigma}.$$

an equation which will be useful.

The Preparation for the Affected Theorem resumed. *Art. No. 27.*

27. In the affected theorem instead of $(x,\, y,\, z)$ we introduce the new coordinates $(\rho,\, \sigma,\, \tau)$. We have

$$J(\phi,\, f,\, \tau) = \frac{d(\phi,\, f,\, \tau)}{d(\rho,\, \sigma,\, \tau)} \cdot \frac{d(\rho,\, \sigma,\, \tau)}{d(x,\, y,\, z)},$$

where the first factor is $= \dfrac{d(\phi,\, f)}{d(\rho,\, \sigma)}$, say that this is $J(\phi,\, f)$, the Jacobian in regard to $\rho,\, \sigma$: and the second factor is at once found to be $= \Delta!$. We have consequently

$$\frac{1}{J(\phi,\, f,\, \tau)} = \frac{1}{\Delta!\, J(\phi,\, f)},$$

and the equation for the affected theorem becomes

$$\Sigma\,(x,\,y,\,z)^{n-3}\,\frac{\delta\phi}{J(\phi,\,f)} = -\,\Delta!\,\left(-\,\frac{\delta\phi_1}{\phi_1}\cdot\frac{\delta\phi_2}{\phi_2}\right),$$

where $(x,\,y,\,z)^{n-3}$ is to be regarded as standing for its value in terms of $(\rho,\,\sigma,\,\tau)$.

Jacobi's Fraction-Theorem. *Art. Nos. 28 to 31.*

28. This is the extension of a well-known theorem, which, in a somewhat disguised form, may be thus written: viz. if U be any rational and integral function $(x,\,1)^m$, then we have

$$\frac{1}{U} = \Sigma\,\frac{1}{x-x'}\,\frac{1}{J(U')},$$

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[book p. 124] or, introducing an arbitrary constant A by the equation $AU = X$, say this is

$$\frac{A}{X}\left(=\frac{1}{U}\right) = \Sigma\,\frac{1}{x-x'}\,\frac{1}{J(U')},$$

where U' is the same function (of x') that U is of x; $J(U')$, $=\dfrac{dU'}{dx'}$, is the Jacobian of U', and the summation extends to all roots x' of the equation $U' = 0$: obviously this is nothing else than the formula for the decomposition of $\dfrac{1}{U}$ into simple fractions.

29. Take now $U = (x,\,y,\,1)^m$, $V = (x,\,y,\,1)^n$ functions of x, y of the degrees m and n respectively, and assume

$$\begin{aligned} AU + BV &= X, \quad \text{a function of } (x,\,1)^{mn}, \\ CU + DV &= Y, \quad \text{,, of } (y,\,1)^{mn}; \end{aligned}$$

viz. let $X = 0$ and $Y = 0$ be the equations obtained by elimination from $U = 0$ and $V = 0$ of the y and the x respectively. The forms are

$$\begin{aligned} A &= (x^{n-1},\,y,\,1)^{mn-m}, \quad & B &= (x^{m-1},\,y,\,1)^{mn-n}, \\ C &= (x,\,y^{n-1},\,1)^{mn-m}, \quad & D &= (x,\,y^{m-1},\,1)^{mn-n}; \end{aligned}$$

where these equations denote the kind of them; that is A is a rational and integral function of the degree $mn-m$ in x and y jointly, but only of the degree $n-1$ in y; and so for the other equations. It follows that

$$AD - BC = (x^{mn-1},\,y^{mn-1},\,1)^{mn+mn-m-n}.$$

The theorem now is

$$\frac{AD-BC}{XY} = \Sigma\,\frac{1}{x-x'\cdot y-y'}\,\frac{1}{J(U',\,V')},$$

where U', V' are the same functions of $(x',\,y')$ that U, V are of $(x,\,y)$; $J(U',\,V')$ is the Jacobian $\dfrac{d(U',\,V')}{d(x',\,y')}$, and the summation extends to all the simultaneous roots x', y' of the equations $U = 0$, $V = 0$.

30. For the proof, observe that $AD - BC$ is a sum of terms of the form $\alpha x^\alpha y^\beta$ where α and β are each of them at most $= mn-1$; hence X being of the degree mn we have $\dfrac{x^\alpha}{X}$ a sum of fractions $\dfrac{L_y}{x - x'}$, where x' is any root of $X = 0$; and similarly $\dfrac{y^\beta}{Y}$ a sum of fractions $\dfrac{M_y}{y - y'}$, where y' is any root of $Y = 0$; multiplying the two expressions and taking into account the several terms $\lambda x^\alpha y^\beta$ of $AD - BC$ we obtain a formula

$$\frac{AD - BC}{XY} = \Sigma \, \frac{K}{x - x' \cdot y - y'},$$

where the summation extends to all the combinations of the mn values of x' with the mn values of y'. But such a formula existing, the coefficients K may be determined.
[book p. 125] in the usual manner, viz. multiplying by XY and then writing $x = x'$, $y = y'$, there is on the right-hand only one term which does not vanish, and we find

$$(AD - BC)_{x'y'} = K\left(\frac{X}{x - x'}\right)_{x'}\left(\frac{Y}{y - y'}\right)_{y'}, \;\; = \; K\left(\frac{dX}{dx}\,\frac{dY}{dy}\right)_{x'y'};$$

where the factor which multiplies K does not vanish.

We distinguish the cases where (x', y') are corresponding or non-corresponding roots of $X = 0$, $Y = 0$, viz. corresponding roots are those for which $U = 0$, $V = 0$, but for non-corresponding roots these equations do not hold good; there are obviously mn pairs of corresponding roots.

In the latter case $(AD - BC)\,U = DX - BY$, $(AD - BC)\,V = -\,CX + AY$, and since for these values $X = 0$, $Y = 0$, the foregoing equation for K gives then $K = 0$.

31. The formula thus is

$$\frac{AD - BC}{XY} = \Sigma \, \frac{K}{x - x' \cdot y - y'},$$

where the summation extends to corresponding roots only; or, for which we have $U = 0$, $V = 0$.
[book p. 126] determinate point, independent of the particular conic through 4, 5 and 6 used for the construction. Thus 4 and 5 being any two chosen points of the cubic, the residual point 6 is, in fact, the intersection with the cubic of the line through 4 and 5.

12. We might go on to the case where the variable curve is a cubic: this meets the fixed curve in 9 points, but they do not determine the variable cubic; for, the cubic being assumed determined by 9 free constants, and corresponding to it the equation gives therefore $\dfrac{d\omega_1}{P_1} + \ldots + \dfrac{d\omega_9}{P_9} = 0$.

The corresponding integral equation is the equation which expresses that the points 1, 2, 3 are in a line, viz. considering y_1, y_2, y_3 as given functions of x_1, x_2, x_3 respectively, this is

$$\begin{vmatrix} x_1, & y_1, & 1 \\ x_2, & y_2, & 1 \\ x_3, & y_3, & 1 \end{vmatrix} = 0,$$

or, in the notation already made use of for such a determinant, $123 = 0$.

32. Reverting to the Cartesian form, we have

$$\frac{xy\,(AD - BC)}{XY} = \Sigma \, \frac{1}{J(U', V')} \, \frac{1}{\left(1 + \dfrac{x'}{x} + \ldots\right)\left(1 + \dfrac{y'}{y} + \ldots\right)};$$

$= \Sigma \, \dfrac{1}{J(U', V')}\left(1 + M\left(\dfrac{x'}{x}, \dfrac{y'}{y}\right) + M_2\left(\dfrac{x'}{x}, \dfrac{y'}{y}\right) + \ldots\right)$; where H_α is the homogeneous sum of the order α of $\dfrac{x'}{x}$, $\dfrac{y'}{y}$, and so $H_1\,(v, w) = v + w$, $H_2\,(v, w) = v^2 + vw + w^2$, &c.

[book p. 127] The left-hand side is

$$(AD-BC)\left(\frac{1}{x^{mn-1}}+\frac{dx}{x^{mn}}+\dots\right)\left(\frac{1}{y^{mn-1}}+\frac{dy}{y^{mn}}+\dots\right),$$

and is $AD-BC$ the terms of the highest order in $(x,\,y)$, say $(AD-BC)_0$, are

$$(AD-BC)_0\,(xy)^{mn-1-1}, = \Sigma\,\frac{1}{J(U',\,V')}\,H^{\alpha+\beta}_{\alpha,\beta\dots}\,(x^{mn-2-\alpha},\,y^{mn-2-\beta}).$$

There is thus on the left-hand no term which is in $(x,\,y)$ of a higher degree than $-(m+n-2)$; hence we the right-hand every term of a higher degree, that is, in $(\alpha,\,\beta)$ a less degree than this, must vanish. Hence the number of these is

$$\theta=\Sigma\,\frac{xy^{\alpha}}{J(U',\,V')},\quad \text{so long as } \alpha+\beta \text{ is } <n-3,$$

where $(x',\,y')$, $J^{n,m-1}$ is the arbitrary function of the degree $n-3$.

But, from the foregoing expression for $(AD-BC)$, it appears that $(AD-BC)_0$ contains the term $px^{mn-2}y^{mn-2}$, and it hence appears that p is the constant term of the quotient $(AD-BC)_0$ divided by $x^{mn-2}y^{mn-2}$, or as this may be written

$$p=\text{const. of } (AD-BC)_0\,(xy)^{2-mn};$$

comparing the two values of p, we have

$$\text{const. of } \frac{(AD-BC)_0\,x^{\alpha}y^{\beta}}{J(U',\,V')}\,(x'y')^{2-m}=\Sigma\,\frac{x^{\alpha}y^{\beta}}{J(U',\,V')}\,(p+q=n-3);$$

and we hence derive

$$\text{const. of } \frac{(AD-BC)_0\,x^{\alpha}y^{\beta}}{J(U',\,V')}\,(x'y')^{2-m}=\Sigma\,\frac{x'^{\alpha}y'^{\beta}}{J(U',\,V')}.$$

[book p. 128] where $(x',y',1)^{m+n-2}$ is the general non-homogeneous function of the degree $m+n-2$, and $(x,y,0)^{m+n-2}$ is obtained from it by attending only to the terms of the highest degree $m+n-2$, and therein substituting x,y for x',y'.

35. We may, it is clear, in the equations for the $(m+n-3)$ and for the $(m+n-2)$ theorems respectively, omit the accents on the right-hand sides; doing this, and moreover in each equation transposing the two sides, the two special theorems are

$$\Sigma\,\frac{(x,y,1)^{m+n-3}}{J(U,V)}=0,\qquad (m+n-3)\text{ theorem},$$

$$\Sigma\,\frac{(x,y,1)^{m+n-2}}{J(U,V)}=\text{const. of } \frac{(AD-BC)_0(x,y,0)^{m+n-2}}{\alpha\beta(xy)^{mn-1}},\qquad (m+n-2)\text{ theorem}.$$

Homogeneous Form of the Special Theorems. *Art. No. 36.*

36. Writing $\frac{x}{z}$, $\frac{y}{z}$ for x, y, and introducing as before the arbitrary linear function $r=ax+by+cz$, we at once obtain, U, V being now homogeneous functions $(x,y,z)^m$ and $(x,y,z)^n$ respectively, and the A, B, C, D being also homogeneous functions accordingly,

$$\Sigma\,\frac{z(x,y,z)^{m+n-3}}{J(U,V)}=0,\qquad (m+n-3)\text{ theorem},$$

$$\Sigma \frac{r(x, y, z)^{m+n-2}}{z\, J(U, V, r)} = \text{const. of } \frac{(AD-BC)_0(x, y, 0)^{m+n-2}}{\alpha\beta(xy)^{mn-1}}, \qquad (m+n-2) \text{ theorem},$$

where the suffix 0 denotes that we are in $AD - BC$ to write $z = 0$.

[book p. 129] The effect of dps of the Curves $U = 0$, $V = 0$. Art. Nos. 37 and 38.

37. We must, in regard to the foregoing special theorems, consider the effect of any dps of the curves $U = 0$, $V = 0$.

Suppose one of the curves, say V, has a dp, but that the other curve U does not pass through it; the dp is not an intersection of U, V, and the theorems are in nowise affected.

If U passes through the dp, then the dp counts twice (say in our case) among the intersections of the curves; the dp is in fact the limit of two close intersections, and the theorems thus becomes negatory.

It, however, the curve ϕ, ϕ^{mn-2} remains finite at (x', y') the dp, then a curve passing through it and a fixed simple intersection of U, V will count as two intersections, and the theorem becomes

$$\Sigma \frac{\phi^{m'n}}{J(U', V')} = 0,$$

where the suffix 0 denotes that we are in $AD - BC$ to retain only the terms of the order

$$2mn - p - q, \text{ viz. in } (\alpha, \beta) \text{ of less than } -(m+n-4)$$

question, which are to be disregarded. And the like in regard to the other theorems

$$\Sigma \frac{(x, y)^{n-3}}{J(U', V')} = 0, \quad (m = n-3),$$

$$\Sigma \frac{(x, y)^{n-3}\, \delta\phi}{J(U', V')} = -\,\text{const.} \frac{(AD-BC)_0\, x^{\alpha} y^{\beta}\, \delta\phi}{(xy)^{mn-2}}, \quad (m = n-2) \text{ theorem};$$

where the suffix 0 denotes that we are in $AD - BC$ to retain only the terms of the order

$$2mn - 4 - p - q.$$

Homogeneous Form of the Special Theorem. *Art. No. 36.*

36. Writing $\frac{x}{z}$, $\frac{y}{z}$ for x, y, and introducing as before the arbitrary linear function $\sigma = \alpha x + \beta y + \gamma z$, and so σ, τ, respectively, we have J, V, V, V becoming J, U, V homogeneous functions accordingly.

If in the last formula we change throughout $\frac{x}{z}$, $\frac{y}{z}$ to x, y, then we have, in the place of $\frac{x'}{z'}$, $\frac{y'}{z'}$ respectively, &c., $\delta\phi$, &c., are functions of the particular function $\frac{x'}{z'}$, $\frac{y'}{z'}$, where x' is any root of $X = 0$, &c.

We have in fact $J(U, V) = z'^{mn-2} J(\phi, f, \tau)$, and similarly for $\frac{1}{xy} \cdot \frac{1}{x'y'}$ in the form $\frac{1}{xy} \cdot \frac{1}{x'y'}$ passing into $\frac{1}{\rho\sigma} \cdot \frac{\Delta f}{\rho\sigma}$, by which means we have, on the right-hand side, Δf in the place of $J(\phi, f)$.

The Pure Theorem—Completion of the Proof. *Art. No. 39.*

39. The theorem was reduced to

$$\Sigma \frac{(x, y, z)^{n-3}\, \delta\phi}{J(\phi, f)} = 0,$$

which is therefore the equation to be proved.

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[book p. 130] The $(m = n-3)$ theorem, writing therein ϕ, f in place of U, V respectively (the degrees being as before m and n), is

$$\Sigma\,\frac{(x,\,y)^{n-3}\,\delta\phi}{J(\phi,\,f)} = 0.$$

Here $(x,\,y)^{n-3}$ is an arbitrary function of the degree $n-3$, and this may therefore be put, $= \Sigma\,\delta(\alpha x + \beta y + \gamma)^{n-3}$, where the α's, β's, γ's are arbitrary, and the variation is in regard only to the δ's; or, since $\phi = 0$ is the equation of an arbitrary curve, it is rather a function of the $(x,\,y)$ coordinates of an arbitrary point on it; the theorem is thus the equation in the same form on the homogeneous variables z, viz. in the homogeneous form

$$\Sigma\,\frac{(x,\,y)^{n-3}\,\delta\phi}{J(\phi,\,f)} = 0.$$

We may write this in the form of an arbitrary function of the degree $n-3$ relating to the homogeneous variables $(x,\,y,\,z)$; transformations to the corresponding form of ϕ, the equation of the variable curve, we have the theorem in its homogeneous form of the highest degree.

The Affected Theorem—Completion of the Proof. **Art. Nos. 40 and 41.**

40. The theorem was reduced to

$$\Sigma\,(x,\,y,\,z)^{n-3}\,\frac{\delta\phi}{J(\phi,\,f)} = -\,\Delta!\,\left(-\,\frac{\delta\phi_1}{\phi_1}\cdot\frac{\delta\phi_2}{\phi_2}\right),$$

which is therefore the equation to be proved.

The $(m = n-2)$ theorem, writing therein ϕ, f in place of U, V respectively, becomes

$$\Sigma\,\frac{(x,\,y)^{n-3}\,\delta\phi}{J(\phi,\,f)} = -\,\text{const.}\,\frac{(AD-BC)_0\,x^{\alpha}y^{\beta}\,\delta\phi}{(xy)^{mn-2}},$$

where, on the right-hand side $(x,\,y)^{n-3}$ is considered as a function of p, q, σ, and we are to put therein $\sigma = 0$; it has been seen (No. 26) that the value is $\dfrac{\Delta f}{\rho\sigma}$, where f is what f, considered as a function of ρ, σ, τ, becomes on writing therein $\tau = 0$; the right-hand side thus becomes

$$= -\,\text{const.}\,\frac{(AD-BC)_0\,\Delta f\,\delta\phi}{\sigma!\,(pq)^{mn-2}};$$